

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

1.a

There are $n + 1$ agents $a_0, \dots, a_n$, with a_0 being Pac-Man and $a_1, \dots, a_n$ the ghosts. A depth is one move from every agent, so d only drops once the last ghost a_n has moved.

$$V_{\text{minimax}}(s, d) = \begin{cases} \text{Utility}(s) & \text{if } \text{IsEnd}(s) \\ \text{Eval}(s) & \text{if } d = 0 \\ \max_{a \in \text{Actions}(s)} V_{\text{minimax}}(\text{Succ}(s, a), d) & \text{if } \text{Player}(s) = a_0 \\ \min_{a \in \text{Actions}(s)} V_{\text{minimax}}(\text{Succ}(s, a), d) & \text{if } \text{Player}(s) = a_i, 1 \leq i < n \\ \min_{a \in \text{Actions}(s)} V_{\text{minimax}}(\text{Succ}(s, a), d - 1) & \text{if } \text{Player}(s) = a_n \end{cases}$$

The first two cases are the two ways to stop: the game has actually ended, or the depth budget has run out and the evaluation function stands in for the rest of the game. The last three recurse, with Pac-Man taking the max and the ghosts the min, and only the last ghost's line passing $d - 1$ down, so that a full round of moves costs one depth rather than one per agent.

3.a

$$V_{\text{exptmax}}(s, d) = \begin{cases} \text{Utility}(s) & \text{if } \text{IsEnd}(s) \\ \text{Eval}(s) & \text{if } d = 0 \\ \max_{a \in \text{Actions}(s)} V_{\text{exptmax}}(\text{Succ}(s, a), d) & \text{if } \text{Player}(s) = a_0 \\ \frac{1}{|\text{Actions}(s)|} \sum_{a \in \text{Actions}(s)} V_{\text{exptmax}}(\text{Succ}(s, a), d) & \text{if } \text{Player}(s) = a_i, 1 \leq i < n \\ \frac{1}{|\text{Actions}(s)|} \sum_{a \in \text{Actions}(s)} V_{\text{exptmax}}(\text{Succ}(s, a), d - 1) & \text{if } \text{Player}(s) = a_n \end{cases}$$

Same as 1a except the ghosts take the average of their successors instead of the min, since each ghost picks every legal move with the same probability $1/|\text{Actions}(s)|$. Pac-Man still takes the max, and d still only drops after the last ghost moves.

5.a

5.b

5.c

6.a

6.b

6.c

6.d

6.e